Class Field Theory VII.9.3

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Original statement

Source `CFT.pdf`, physical page 233.

Lemma 9.3 (Key case of the Existence Theorem) *Let K be a number field containing a primitive pth root of 1 (p prime). Then every open subgroup V of $\boldsymbol{C}_K$ such that $\boldsymbol{C}_K/V$ is a finite group killed by p is a norm group.*

Original proof

Source `CFT.pdf`, physical pages 233–234.

Proof. Let $S \supset S_\infty$ be a finite set of primes of K containing the infinite primes, those dividing p, and enough primes so that $\mathbb{I}_K = K^\times \cdot \mathbb{I}_S$. Let L be the extension of K corresponding by Kummer theory to the group $U(S) \cdot K^{\times p}$, i.e., $L = K[U(S)^{\frac{1}{p}}]$, and let

$$E = \prod_{v \in S} K_v^{\times p} \times \prod_{v \notin S} U_v.$$

We shall prove that $K^\times \cdot E = K^\times \cdot \mathrm{Nm}(\mathbb{I}_L)$ by verifying that

(a) $E \subset \mathrm{Nm}(\mathbb{I}_L)$;

(b) $(\mathbb{I}_K : K^\times \cdot E) = p^s = (\mathbb{I}_K : K^\times \cdot \mathrm{Nm}_{L/K}(\mathbb{I}_L))$.

For any prime v of K and prime w of L lying over it, the local Artin map is an isomorphism

$$K_v^\times / \mathrm{Nm}(L_w^\times) \longrightarrow \mathrm{Gal}(L_w/K_v).$$

Because L/K is has exponent p, $\mathrm{Nm}(L_w^\times) \supset K_v^{\times p}$.

For any prime $v \notin S$, L is unramified over K and v, and so the norm map $U_w \longrightarrow U_v$ is onto.

On combining the statements in the last two paragraphs, we obtain (a).

From the Reciprocity Law,

$$(\mathbb{I}_K : K^\times \cdot \mathrm{Nm}(\mathbb{I}_L)) = [L : K],$$

and from Kummer theory,

$$[L : K] = (U(S) \cdot K^{\times p} : K^{\times p}).$$

But

$$U(S) \cdot K^{\times p}/K^{\times p} \approx U(S)/U(S) \cap K^{\times p}.$$

If $a^p \in U(S)$, then $a \in U(S)$, and so $U(S) \cap K^{\times p} = U(S)^p$. Now, by the Dirichlet Unit Theorem (ANT, 5.9),

$$U(S) \approx U(S)_{\text{torsion}} \oplus \mathbb{Z}^{s-1},$$

and $U(S)_{\text{torsion}}$ is the group of roots of 1 in K, which is a cyclic group whose order is divisible by p. Hence $(U(S) : U(S)^p) = p^s$.

On the other hand,

$$(\mathbb{I}_K : K^\times \cdot E) = (\mathbb{I}_S \cdot K^\times : E \cdot K^\times),$$

which, by (6.5), equals

$$(\mathbb{I}_S : E)/(\mathbb{I}_S \cap K^\times : E \cap K^\times).$$

Therefore (see 6.8),

$$(\mathbb{I}_S : E) = \prod_{v \in S} (K_v^\times : K_v^{\times p}) = \prod_{v \in S} \frac{p}{|p|_v} p = p^{2s}.$$

Here, we have used that K contains a primitive pth root of 1 and that S contains all v for which $|p|_v \neq 1$, and so $\prod_{v \in S} |p|_v = \prod_{\text{all } v} |p|_v = 1$ by the product formula. It follows that $K^\times \cdot E = K^\times \cdot \operatorname{Nm} \mathbb{I}_L$.

Now let V be an open subgroup of $\boldsymbol{C}_K$ such that $\boldsymbol{C}_K/V$ is killed by p, and let U be the inverse image of V in $\mathbb{I}_K$. Then U is open in $\mathbb{I}_K$ and so there is a finite set of primes S such that $U \supset \prod_{v \in S} 1 \times \prod_{v \notin S} U_v$. Moreover, $\mathbb{I}_K/U$ has exponent p, and so $U \supset \mathbb{I}_K^p$. Hence $U \supset E \cdot K^\times$, and because $E \cdot K^\times/K^\times$ is a norm group, so also must be $U/K^\times = V$. □

Original supporting excerpts

Source `CFT.pdf`, physical pages 232–233.

LEMMA 9.1 *If U is a norm group, and $V \supset U$, then V also is a norm group.*
PROOF. Suppose $U = \operatorname{Nm} \boldsymbol{C}_L$. According to the Reciprocity Law, the Artin map defines an isomorphism

$$\phi : \boldsymbol{C}_K/U \longrightarrow \operatorname{Gal}(L/K).$$

If M is the fixed field of $\phi(V)$, then ϕ defines an isomorphism

$$\boldsymbol{C}_K/V \longrightarrow \operatorname{Gal}(M/K),$$

but, according to the Reciprocity Law, the kernel of $\phi : \boldsymbol{C} \longrightarrow \operatorname{Gal}(M/K)$ is $\operatorname{Nm}_{M/K} \boldsymbol{C}_M$. □

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PROPOSITION 6.8 *Let K be a local field of characteristic zero, and let μ_n be the group of nth roots of 1 in $K^\times$. Then*

$$(K^\times : K^{\times n}) = n \frac{|\mu_n|}{|n|}.$$

If K is nonarchimedean and U is the group of units in K, then

$$(U : U^n) = \frac{|\mu_n|}{|n|}.$$

PROOF. If $K = \mathbb{C}$, the first equation becomes $1 = n\frac{n}{n^2}$, and if $K = \mathbb{R}$, it becomes $1 = n\frac{1}{n}$ when n is odd and $2 = n\frac{2}{n}$ when n is even. If K is nonarchimedean, $K^\times \approx U \times \mathbb{Z}$, and so

$$(K^\times : K^{\times n}) = (U : U^n)(\mathbb{Z} : n\mathbb{Z}) = (U : U^n)n.$$

Therefore, it remains to prove the second equation.

For an abelian group M, we write

$$h_n(M) = (M : nM)/(M_n : 1), \qquad M_n = \{x \in M \mid nx = 0\}.$$

Then $h_n(M)$ is the Herbrand quotient of M regarded as a $\mathbb{Z}/n\mathbb{Z}$-module with trivial action, and so we may apply the results in II, §2.

As we saw in the proof of (III, 2.4), the exponential map defines an isomorphism from a subgroup of finite index in $\mathcal{O}_K$ onto a subgroup of finite index in U. Therefore

$$h_n(U) = h_n(\mathcal{O}_K) = (\mathcal{O}_K : n\mathcal{O}_K) \stackrel{\text{def}}{=} |n|^{-1}.$$

Hence

$$(U : U^n) = \frac{(U_n : 1)}{|n|}.$$

As $U_n = \mu_n$, this proves the second equation. □

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THEOREM A.3 *The map*

$$L \longmapsto K^\times \cap L^{\times n}/K^{\times n}$$

defines a one-to-one correspondence between the finite abelian extensions of K of exponent n contained in some fixed algebraic closure Ω of K and the finite subgroups B of $K^\times/K^{\times n}$. The extension corresponding to B is $K[B^{\frac{1}{n}}]$, the smallest subfield of Ω containing K and an nth root of each element of B. If $L \leftrightarrow B$, then $[L : K] = (B : K^{\times n})$.
PROOF. See FT 5.29. □

Adjacent original result for the index comparison

Source `CFT.pdf`*, physical page 233. This excerpt is separate from the selected proof of Lemma 9.3.*

PROPOSITION 9.2 *Let K be a number field containing a primitive nth root of 1, and let $S \supset S_\infty$ be a finite set of primes of K containing all those dividing n and enough primes to generate the class group of K. Any $a \in K^\times$ such that*

- *a is an nth power in K_v for all $v \in S$;*
- *a is a unit in K_v for all $v \notin S$.*

is an nth power in K.
PROOF.[7] Let $L = K[a^{1/n}]$—because $\zeta_n \in K$, this is an abelian extension of K. For every prime $v \in S$, $X^n - a$ splits completely in $K_v[X]$, and so $L_w = K_v$ for all $w \mid v$. Hence the norm map $L_w^\times \longrightarrow K_v^\times$ is onto. On the other hand, L is unramified over K at any prime $v \notin S$, and so the norm map $\mathrm{Nm}_{L/K} : U_w \longrightarrow U_v$ is onto. Therefore, $\mathrm{Nm}_{L/K}(\mathbb{I}_L) \supset \mathbb{I}_S$, and so

$$K^\times \cdot \mathrm{Nm}_{L/K}(\mathbb{I}_L) \supset K^\times \cdot \mathbb{I}_S = \mathbb{I}_K.$$

The Reciprocity Law now shows that $L = K$, and so a is an nth power in K. □

[7]From Wojtek Wawrów: Since $L = K$ in the proof is cyclic, it appears that the proof only uses the First Inequality (specifically Lemma 4.5), not the full Reciprocity Law. Regardless, it appears that this Proposition is not used anywhere.